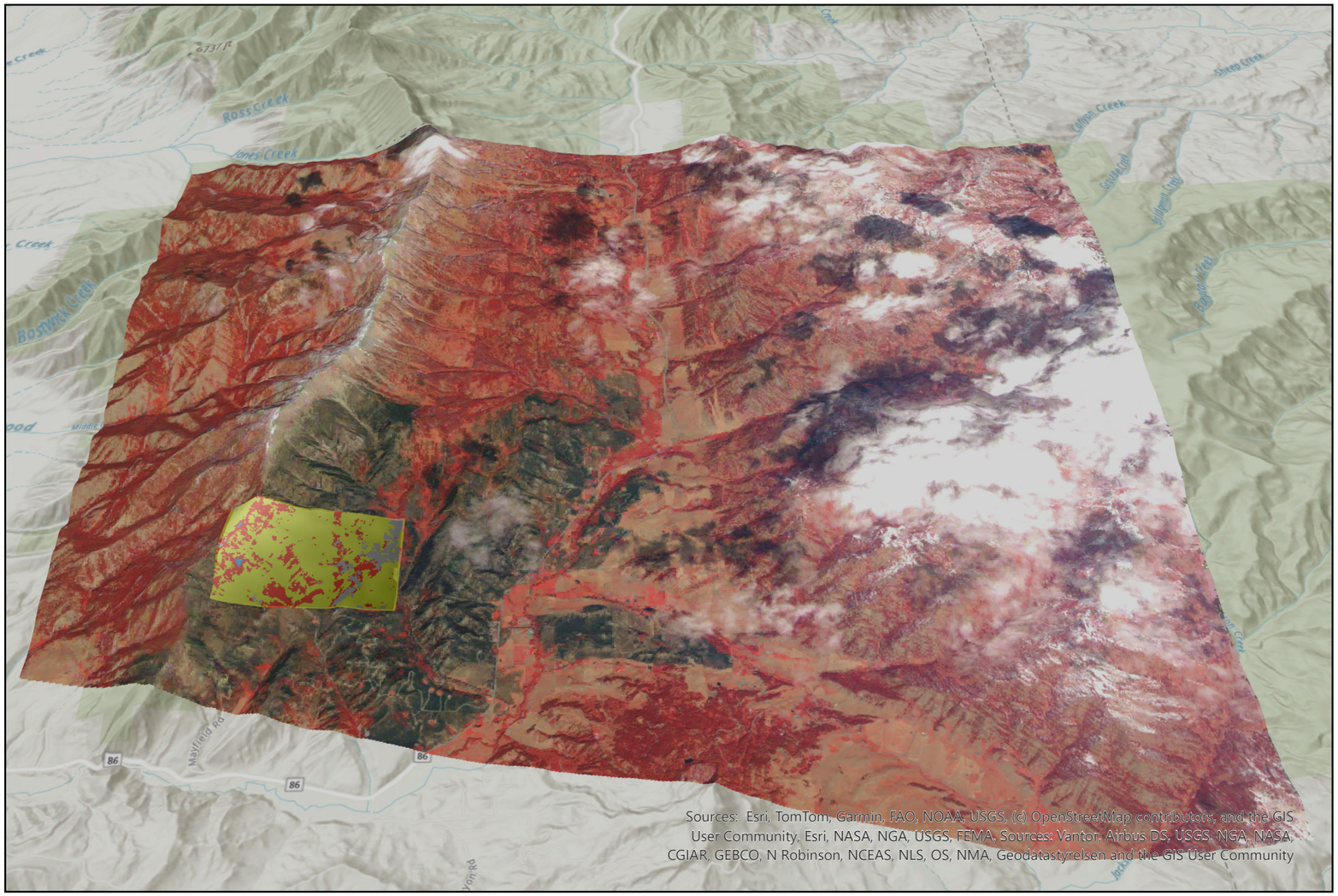
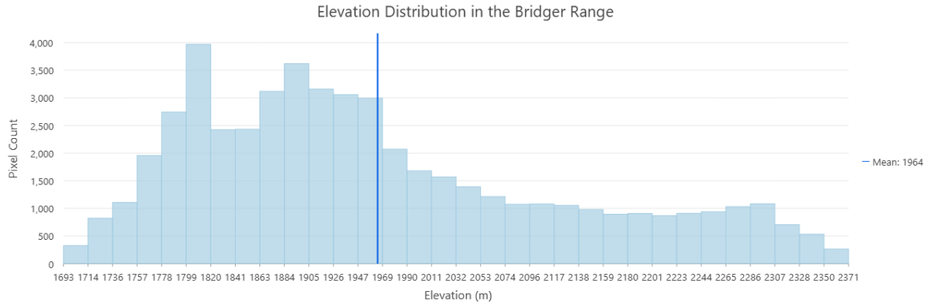
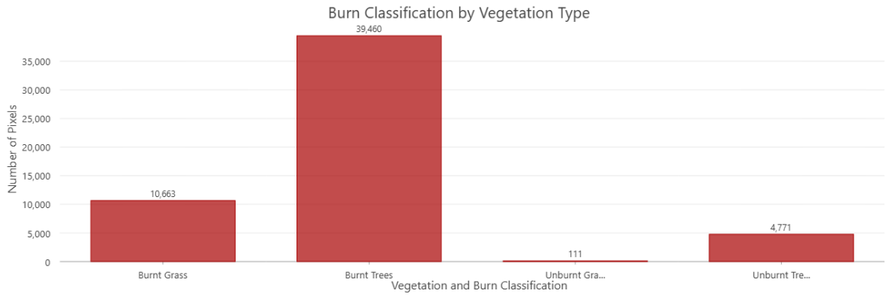
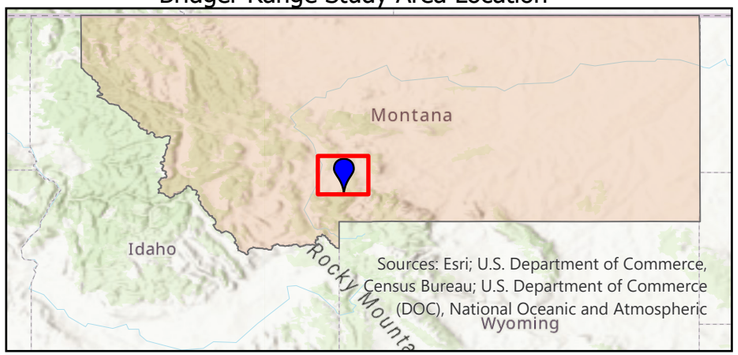
Wildfire Burn Patterns and Elevation in the Bridger Range, Montana









N

0 500 1,000 Kilometers

Burn Classification

- Unburnt Trees
- Burnt Trees
- Unburnt Grass
- Burnt Grass

Interpretation of Results

The elevation histogram shows that elevations within the study area range from approximately 1,693 to 2,371 m, with a mean of 1,964 m. Lower and intermediate elevations contain more raster cells than the highest elevations, indicating an uneven distribution of terrain elevations across the study area.

The burn classification indicates that burned trees account for the largest number of classified pixels (39,460), followed by burned grass (10,663). Unburned trees and unburned grass account for 4,771 and 111 pixels, respectively. These results suggest that fire impacts were particularly extensive in areas classified as forest vegetation.

Data Sources and Projection

Satellite Imagery: Planet four-band imagery (BridgBurn090920), provided through GPHY 504 Lab 10.
Burn Classification: ClassBrigBurn, supplied with the GPHY 504 laboratory dataset; processed to create Burn focus.
Elevation: DEM Broad and DEM focus, supplied with the GPHY 504 laboratory dataset.
Basemap: Esri World Topographic Map and World Hillshade.
CRS: WGS 1984 UTM Zone 12N (EPSG: 32612).
Map Projection: Transverse Mercator (UTM Zone 12N)
Software: ArcGIS Pro.

Map by: Omotayo Ajao